The simple form of the Radar Equation

If the transmitter power P_t is radiated by an isotropic antenna (one that radiates uniformly in all directions), the power density (watt/Area) at a distance R from the radar is equal to the radiated power divided by the surface area $4\pi R^2$ of an imaginary sphere of radius 'R'.

$\therefore$ power density at Range R from an isotropic antenna $= \dfrac{P_t}{4\pi R^2}$

The power density at the target from a directive antenna with a transmitting Gain 'G' is then

Power density at Range R from a directive antenna $= \dfrac{P_t G}{4\pi R^2}$

The Radar cross section of the target determines the power density returned to the radar for a particular power density incident on the target. It is denoted by σ

Reradiated power density back at the radar $= \dfrac{P_t G}{4\pi R^2} \cdot \dfrac{\sigma}{4\pi R^2}$

The radar antenna captures a portion of the echo energy incident on it. If the effective area of the receiving antenna is 'Ae' and the power P_r received by radar is

$$P_r = \frac{P_t G}{4\pi R^2} \frac{\sigma}{4\pi R^2} A_e$$

$$= \frac{P_t G \sigma A_e}{(4\pi R^2)^2}$$

The maximum Radar range (R_{max}) is the distance beyond which the target can't be detected. It occurs when the received echo signal power P_r just equals the minimum detectable signal S_{min}

$$(R_{max})^4 = \frac{P_t G \sigma A_e}{(4\pi)^2 S_{min}}$$

$$\therefore R_{max} = \left(\frac{P_t G \sigma A_e}{(4\pi)^2 S_{min}}\right)^{1/4}$$

This eqn is called fundamental radar eqn.

* $G = \frac{4\pi A_e}{\lambda^2}$, $A_e = \frac{G\lambda^2}{4\pi}$

$$\therefore R_{max} = \left(\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 S_{min}}\right)^{1/4}$$

→ In a Radar system it is observed that the echo received is after 9.15 μsec. calculate the distance of the stationary object from the radar.

Sol

$$T_R = 9.15\,\mu\,sec$$

$$R = \frac{C\,T_R}{2}$$

$$= \frac{3\times10^{8}\ m/sec \times 9.15\times10^{-6}\ sec}{2}$$

$$= 1.372\ Km$$

→ A radar system transmits pulses of duration 2μs and repetition rate of 1KHz. Find the maximum and minimum range of radar.

Sol

$$PRF\ f_r = 1\,KHz$$

$$P.W = 2\mu sec.$$

The maximum Range of radar is

$$R_{max} = \frac{c}{2f_r} = \frac{3\times10^{8}\ m/s}{2\times1\times10^{3}}$$

$$= 1.5\times10^{5}\ m.$$

$$R_{min} = \frac{c.\,Pw}{2} = \frac{3\times10^{8}\times2\times10^{-6}}{2}$$

$$= 300\ meters.$$

→ what is the duty cycle of radar with a pulse width of 4 μsec and a pulse repetition time of 8 msec?

Sol

$$\text{Duty cycle} = \frac{Pw}{PRT} = \frac{4\times10^{-6}}{8\times10^{-3}} = 0.0005.$$

→ A Radar is to have a maximum of 300km. what is the maximum allowable pulse repetition freq for unambiguous reception?

Sol

The maximum unambiguous range is given by

$$R_{max} = \frac{c}{2f_r}$$

$$f_r = \frac{c}{2R_{max}}$$

$$= \frac{3\times10^8}{2\times300\times10^3}$$

$= 500$ pulse per sec

$= 0.5\,KHz$

→ calculate the maximum range of radar which operates at a freq of 10GHz, peak pulse power of 600Kw. If the antenna effective area is $5m^2$ and the area of target is $20m^2$, minimum receivable power is 10^{-13} watt.

Sol

$P_t = 600\,Kw$ $\quad \sigma = 20m^2$ $\quad A_e = 5m^2$ $\quad P_{min}$ (or) $S_{min} = 10^{-13}\,w$

$f = 10\times10^9\,Hz$

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{10\times10^9} = 0.03\,m$$

$$G_t = \frac{4\pi A_e}{\lambda^2} = 69.813\times10^3$$

$$R_{max} = \left(\frac{P_t G A_e \sigma}{(4\pi)^2 P_{min}}\right)^{1/4}$$

$= 717\,Km$

→ A marine radar operating at 10 GHz has a max. range of 50 Km with an antenn gain of 4000. If the transmitter has a power of 250 KW and and minimum detectable signal of 10^{-11} W. Determine the cross-sec of the target the radar can sight.

<u>sol</u>

$f = 10\times10^9$ Hz $\quad R_{max} = 50$ Km

$P_t = 250$ KW $\quad G_t = 4000$

P_{min} (or) $S_{min} = 10^{-11}$ W

$$\lambda = \frac{c}{f} = 0.03\,m$$

$$\therefore R_{max} = \left[\frac{P_t G_t^2 \lambda^2 \sigma}{(4\pi)^3 S_{min}}\right]^{1/4}$$

$$\sigma = \frac{R_{max}^4 (4\pi)^3 S_{min}}{P_t G_t^2 \lambda^2}$$

$$= 34.45\ m^2$$

→ An S-band radar transmitting at 3 GHz radiates 200 KW. Determine the signal power density at ranges 100 nautical miles if the effective area of the radar antenna is 9 m².

<u>sol</u>

$f = 3$ GHz $= 3\times10^9$ Hz

$P_t = 200$ KW $= 200\times10^3$ watts

$R = 100$ nautical miles

1 nautical mile $= 1.8412\times10^3$ m

$\therefore R = 1.8412\times10^5$ meter

$A_e = 9\,m^2$ $\qquad \lambda = \frac{c}{f} = \frac{3\times10^8}{3\times10^9} = 0.1\,m$

$$G_t = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi\times9}{(0.1)^2} = 11.3\times10^3$$

Power density by directive antenna is

$$P = \frac{P_t G_t}{4\pi R^2}$$

$$= 5.248\ mW/m^2$$

→ A Radar operating at 3GHz radiating Power of 200 KW. Calculate the power of the reflected signal at the radar with a $20m^2$ target at 300 nautical miles: Take $A_e = 9m^2$

Sol

$$P_r = \frac{P_t G_t A_e \sigma}{(4\pi)^2 R^4} \qquad G_t = \frac{4\pi A_e}{\lambda^2}$$

$$R = 300\times1.8412\times10^3\ m$$

$$= 27.034\times10^{-15}\ \text{watts}$$

→ Find the maximum range of a radar, the transmitted power is 250KW, cross sectional area of the target is $12.5\,m^2$, minimum power received is 10^{-3} watt. receiver antenna gain is 2000 and operating wavelength 16cm.

Sol

$$R_{max} = \left(\frac{P_t G_t^2 \lambda^2 \sigma}{(4\pi)^3 S_{min}}\right)^{1/4}$$

$G_r = G_t$
common antenna is used for transmission & reception
$\therefore G_r = G_t$

$$= 200.39\ Km$$

Scanned by CamScanner

→ A pulsed radar operating at 10GHz has an antenna with a gain of 28dB and a transmitting power of 2KW. If it is defined to detect a target with cross section of 12 m^2 and the minimum detectable signal $S_{min} = -90dBm$. what is the maximum range of radar.

Sol

$f = 10 \times 10^9$ Hz $\quad \lambda = \frac{c}{f} = 0.03m$

$G_t = 28dB = 630.95 \quad P_t = 2 \times 10^3 w$

$\sigma = 12\ m^2 \quad P_{min}$ (or) $S_{min} = -90\ dBm$

$= 10^{-12}$ watt

$$\therefore R_{max} = \left(\frac{P_t G_t \lambda^2 \sigma}{(4\pi)^3 S_{min}} \right)^{1/4}$$

$= 1619 = 1.619$ km.

Limitation of Radar Range eqn

$$R_{max} = \left(\frac{P_t G A_e \sigma}{(4\pi)^2 S_{min}} \right)^{1/4} \quad \text{—(1)}$$

$G = \frac{4\pi A_e}{\lambda^2}$

$$R_{max} = \left(\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 S_{min}} \right)^{1/4} \quad \text{—(2)}$$

$A_e = \frac{G\lambda^2}{4\pi}$

$$R_{max} = \left(\frac{P_t A_e^2 \sigma}{4\pi \lambda^2 S_{min}} \right)^{1/4} \quad \text{—(3)}$$

These three forms of the radar eqns are basically the same but there are diff in interpretation.

for example in eq (2)

$$R_{max} \propto \lambda^{1/2}$$

& in eq (3)

$$R_{max} \propto \lambda^{-1/2}$$ which is just opposite.

The correct interpretation depends on whether the antenna gain is held constant with change in wavelength or freq as implied by eq (2)

(or) the effective area is held constant for eq (3)

Receiver Noise

Noise is unwanted electromagnetic energy which interferes with the ability of the receiver to detect the wanted signal thus limiting the receiver sensitivity. It may originate within the receiver itself or it may enter via the receiving antenna along with the desired signal.

If the radar were to operate in a perfectly noise free environment so that no external noise accompany the target signal, & if the receiver itself were so perfect that it did not generate any excess noise, there would be still be noise generated by the thermal motion of the conduction electrons in the ohmic portion of the receiver i/p statges. This is called thermal noise or Johnson noise.

The available thermal noise power generated by a receiver of B.W B_n (hertz) at a temparature T (°K) is

thermal noise power $= KTB_n$

K = Boltzmann's constant $= 1.38 \times 10^{-23}$ J/deg.

~~The~~ For the superhetrodyne receiver mostly used in radar, the receiver B.W is approximately that of the intermediate freq B.W. (I.F amplifier)

$$B_n = \frac{\int_{-\infty}^{\infty} |H(f)|^2 \, df}{|H(f_0)|^2}$$

$H(f)$ = freq response fun of the IF amplifier

f_0 = freq of maximum response.

The Bandwidth B_n is called the noise bandwidth is the bandwidth of the equivalent rectangular filter whose noise power o/p is same as the filter with freq response fun $H(f)$.

But noise bandwidth is not theoretically same as the 3-dB (or) half power bandwidth. 3-dB BW is defined by the separation b/w the points of the freq response fun $H(f)$ where the response is reduced 0.707 (3dB in power) from its maximum value.

Actually 3dB - bandwidth is used in many cases as an approximation to the noise B.W.

The measure of the noise out of a real receiver to that from the ideal receiver with only thermal noise is called the noise figure and is defined as

$$F_n = \frac{N_{out}}{K T_0 B_n G_a} = \frac{\text{Noise out of practical receiver}}{\text{Noise out of ideal receiver at std temp } T_0}$$

G_a = available gain

T_0 = 290 K std temp

The available gain G_a is the ratio of the signal S_{out} to the signal in S_{in}, with both o/p and i/p matched to deliver maximum o/p power.

The i/p Noise N_{in}, in an ideal receiver is $K T_0 B_n$

$$\therefore \text{Noise figure } F_n = \frac{N_{out}}{\frac{S_{out}}{S_{in}} N_{in}}$$

$$= \frac{S_{in}/N_{in}}{S_{out}/N_{out}}$$

$\therefore$ i/p signal $\quad S_{in} = \frac{K T_0 B_n F_n S_o}{N_o}$

If the minimum detectable signal S_{min} is that value of S_{in}, which corresponds to the min detectable signal to noise at the o/p of the IF $(S_{out}/N_{out})_{min}$.

$$S_{min} = K T_0 B_n F_n \left(\frac{S_{out}}{N_{out}}\right)_{min}$$

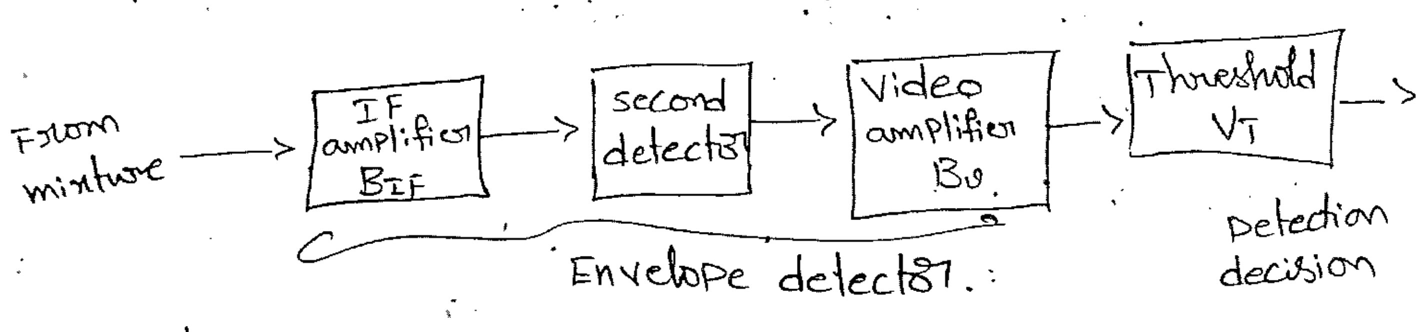
∴ the modified Radar eq^n is

$$R^4_{max} = \frac{P_t G A_e \sigma}{(4\pi)^2 K T_0 B_n F_n (S_0/N_0)_{min}}$$

Signal to noise Ratio

Signal to noise ratio is very important as far as radar is concerned. Because presence of target or not have small diff. Statistical noise theory will be applied to obtain S/N at the o/p of the IF amplifier necessary to achieve a specified prob of detection without exceeding a specified prob of false alarm.

The Details of system that is considered



Conside

-> IF amplifier with Band width B_{IF} followed by a second detector and a video amplifier with Band width B_V.

-> the second detector and video amplifier are assumed to form an envelope detector, that is one which rejects the carrier freq. but passes the modulation envelope.

→ To extract the modulation envelope, the video Bw must be wide enough to pass the low freq components generated by the second detector but no so wide as to pass the high freq components at or near the IF.

→ The video bandwidth B_v must be greater than $B_{IF}/2$ in order to pass all video modulation.

Probability of false alarm

The receiver noise at the i/p to the IF filter is described by the gaussian probability density fun with mean value zero

$$P(v) = \frac{1}{\sqrt{2\pi \psi_0}} \exp \frac{-v^2}{2\psi_0}$$

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-a_x)^2}{2\sigma^2}}$$

S.O. Rice has shown that when the gaussian noise is passed through the IF filter, the prob density fun of the envelope R is given by a form of the Rayleigh pdf.

Rayleigh $P(x) = \frac{2}{b}(x-a)\, e^{-(x-a)^2/b}$

$$P(R) = \frac{R}{\psi_0} \exp\left(\frac{-R^2}{2\psi_0}\right)$$

R → amplitude of envelope of the filter

The prob that the envelope of the noise voltage will exceed the voltage threshold V_T

$$P(V_T < R < \infty) = \int_{V_T}^{\infty} \frac{R}{\psi_0} \exp\left(\frac{-R^2}{2\psi_0}\right) dR$$

$$= \exp\left(\frac{-V_T^2}{2\psi_0}\right) = P_{fa}$$

whenever the voltage envelope exceeds the threshold V_T, a target is considered to have been detected, since the prob of a false alarm is the prob that noise will cross the threshold.

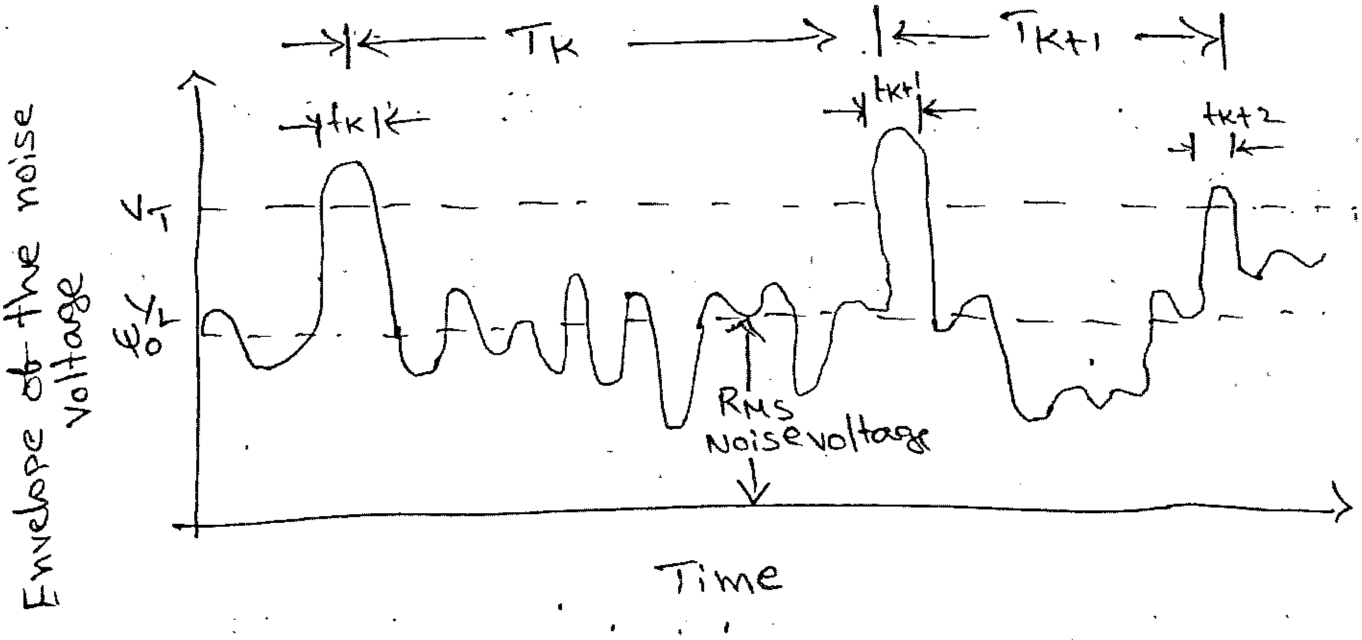
∴ prob of false alarm

$$P_{fa} = \exp\left(\frac{-V_T^2}{2\psi_0}\right)$$

The time b/w false alarms T_{fa} is also a better measure of the effect of Noise on the radar performance. The average time b/w crossings of the threshold by noise alone is defined as the false alarm time T_{fa}.

$$T_{fa} = \lim_{N \to \infty} \frac{1}{N} \sum_{K=1}^{N} T_K$$

T_K is the time b/w crossings of the threshold V_T by the noise envelope.



The false alarm probability may also be defined as the ratio of the duration of time the envelope is actually above the threshold to the total time it could have been above the threshold

i.e;

$$P_{fa} = \frac{\sum_{K=1}^{N} t_K}{\sum_{K=1}^{N} T_K} = \frac{(t_K)_{av}}{(T_K)_{av}} = \frac{1}{T_{fa} B}$$

The avg duration of a threshold crossing by noise $(t_K)_{avg}$ is approximately the reciprocal of the IF Bandwidth B. The avg of T_K is the false alarm time T_{fa}.

$$T_{fa} = \frac{1}{B} \exp\left(\frac{V_T^2}{2\psi_0}\right)$$

Prob of detection

So far we have discussed only the noise i/p at the radar receiver. Now consider an echo signal represented as a sine wave of amplitude A along with gaussian noise at the i/p of the envelope detector.

∴ The prob density fun of the envelope R at the o/p is given by

$$P_s(R) = \frac{R}{\psi_0} \exp\left(-\frac{R^2+A^2}{2\psi_0}\right) I_0\left(\frac{RA}{\psi_0}\right) \quad \text{—①}$$

where $I_0(Z)$ is the modified Bessel fun of zero order and argument Z.

For large Z, an asymptotic expansion for $I_0(Z)$ is

$$I_0(Z) = \frac{e^Z}{\sqrt{2\pi Z}}\left(1+\frac{1}{8Z}+\cdots\right)$$

when the signal is absent A=0 eq (1) reduced to

then $$P(R) = \frac{R}{\psi_0}\exp\left(\frac{-R^2}{2\psi_0}\right)$$

The prob of detecting the signal is the prob that the envelope R will exceed the threshold V_T.

$\therefore$ The prob of detection is

$$P_d = \int_{V_T}^{\infty} P_s(R)\, dR$$

$$= \int_{V_T}^{\infty} \frac{R}{\psi_0}\exp\left[\frac{-(R^2+A^2)}{2\psi_0}\right] I_0\left(\frac{RA}{\psi_0}\right) dR$$

In radar systems analysis it is more convenient to use signal to noise power ratio S/N than $A^2/2\psi_0$

$$\frac{A^2}{\psi_0^{1/2}} = \frac{\text{signal amplitude}}{\text{rms noise voltage}} = \frac{\sqrt{2}\,(\text{rms signal voltage})}{\text{r.m.s noise voltage}}$$

$$= \left(2\,\frac{\text{signal power}}{\text{noise power}}\right)^{1/2} = \left(\frac{2S}{N}\right)^{1/2}$$

Integration of Radar Pulses

The no. of pulses returned after hitting target is given by

$$n = \frac{\theta_B f_P}{\theta_S} = \frac{\theta_B f_P}{6 w_r}$$

θ_B = antenna Beam width deg

f_P = pulse repetition freq Hz

θ_S = antenna scanning rate deg/sec

w_r = revolutions per minute (rpm) if 360° rotating antenna.

The process of summing all the radar echos available from a target is called integration.

If the integration of pulses is performed before second detector in radar receiver is known as predetection integration or coherent integration. after second detector is called post detection integration or non coherent integration.

→ If n pulses of sam S/N were perfectly integrated by an ideal lossless predetection integrator, the integrated S/N will be exactly n times that of a single pulse

$$\therefore \left(\frac{S}{N}\right)_n = \frac{\left(\frac{S}{N}\right)_1}{n}$$

→ If n pulses of same S/N are integrated by post detection integrator, the integrated S/N will be less than n time of single pulse.

This loss in integration efficiency is caused by the non linear action of the second dector, which converts some of the signal energy to noise energy in the rectification process.

The integration efficiency for post-detection integration is given by

$$E_i(n) = \frac{(S/N)_1}{n(S/N)_n} \quad — ①$$

The improvement in S/N ratio when n pulses are integrated is called the "integration improvement factor" $I_i(n) = n E_i(n)$.

Marcum defined an integration loss in dB as $L_i(n) = 10 \log [1/E_i(n)]$.

The radar eqn with n pulses integrated can be written as

$$R_{max}^4 = \frac{P_t G A_e \sigma}{(4\pi)^2 K T_0 B_n F_n (S/N)_n}$$

from eq ①

$$R_{max}^4 = \frac{P_t G A_e \sigma \, n \, E_i(n)}{(4\pi)^2 K T_0 B_n F_n (S/N)_1}$$

Radar cross-section of targets

A radar cross section is defined as the ratio of its effective isotropic scattered power to the incident power density.

$$\sigma = \frac{\text{Power reflected towards source / unit solid angle}}{\text{incident power density} / 4\pi}$$

$$= \lim_{R \to \infty} 4\pi R^2 \left|\frac{E_r}{E_i}\right|^2$$

R is the range of the target

E_r is the strength of reflected field at radar

E_i is the " incident at target

→ The radar cross-section depends on the characteristic dimensions of the object compared to the radar wavelength.

(i) when the wavelength is large compared to the object's dimensions, scattering is said to be in Rayleigh region

(ii) when the wavelength is small compared to the object's dimensions is called the optical region.

(iii) In b/w the Rayleigh and optical regions is the resonance region where the radar wavelength is comparable to the objects dimensions

For many objects the radar cross section is larger in the resonance region than in the other two regions.

Transmitter Power

$$R_{max} = \left(\frac{P_t G A_e \sigma}{(4\pi)^2 S_{min}} \right)^{1/4} \quad -(1)$$

the power P_t in the eq (1) is also called peak power of the pulse. The avg power P_{avg} of a radar is also an important measure of radar performance than the peak power. It is defined as the average transmitter power over the duration of the total transmission.

If the transmitter waveform is a train of rectangular pulses of width τ and constant pulse repetition period $T_p = 1/f_p$ then the avg power is related to peak power by

$$P_{av} = \frac{P_t \tau}{T_p} = P_t \tau f_p$$

The radar duty cycle $= \frac{P_{avg}}{P_t} = \tau f_p$

$$\therefore R^4_{max} = \frac{P_{av} G A_e \sigma n E_i(n)}{(4\pi)^2 k T_0 F_n (B_n \tau) f_p (S/N)_1}$$

If the transmitted waveform is not rectangular pulse, sometimes it is more convenient to express radar eq^n in terms of energy

The energy per pulse

$$E_p = P_t \tau = \frac{P_{av}}{f_p}$$

$$\therefore R_{max}^4 = \frac{E_p G A_e \sigma \, n \, E_i(n)}{(4\pi)^2 k T_0 F_n (B_n \tau)(S/N)_1}$$

$$= \frac{E_T G A_e \sigma \, E_i(n)}{(4\pi)^2 k T_0 F_n (B_n \tau)(S/N)_1}$$

where E_T is the total energy of the 'n' pulses which equals nE_p i.e; $E_T = nE_p$

The bandwidth and the pulse width are grouped together since the product of two is approximately unity in a well designed radar.